

A Release-Native DR11-South Strong-Lens Search: the CLAUDENET v4 Fine-Tuned Re-Sweep

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Abstract

This report is the CLAUDENET v4 follow-on to the as-run DR11-south campaign (the sibling report, which swept and vetted with v3blend8 against HSC PDR3 (Aihara et al., 2022) and SuGOHI (Sonnenfeld et al., 2018)). It documents two release-portable levers and the candidate catalogue they produce. (i) The apparent DR9→DR11 grade-A recall collapse (62%→41% at the 10^{-4} operating point) is an *operating-point* artifact, not a model failure: the five-member calibrated *mean* separates DR11 lenses from random DR11 galaxies at AUC 0.9955, and replacing the per-member union with the mean recovers grade-A recall to 75% at the same survivor budget. (ii) A *release-native warm-start fine-tune* of the three swappable members — on a $\sim 13\times$ -expanded native positive pool (3,171 net-new confirmed lenses) plus DR11-native hard negatives — passes a held-out gate, lifting Storfer grade-A recall $0.544 \rightarrow 0.796$ (the lrg+companion hard residual the resolution-limited v3blend8 could not crack). Re-sweeping all 5.38×10^7 DR11-south galaxies with the fine-tuned v4 ensemble and selecting the top 150k by mean yields the v4 candidate set, whose held-out recall (Storfer/Inchausti, training-excluded) reaches **0.87** (Inchausti grade-A) and **0.825** (Storfer grade-A) — the best of any configuration, and a substantially new pool (+134k candidates beyond the union-95k). The v4 catalogue is recall-richer but its new candidates are *pending* LENSJUDGE HSC tier-2 vetting (§4, in progress); the completed, HSC-vetted catalogue remains the v3blend8 as-run set of the sibling report.

1 Premise: a release-native re-sweep

The as-run DR11-south campaign (sibling report) established the resolution-limited thesis with v3blend8: a DESI pre-screen, HSC tier-2 confirmation in the overlap, 24 HSC-graded A/B (mostly SuGOHI cross-matches), and an honest “no net-new lens from DECaLS alone.” It also surfaced — but did not deploy — the diagnosis that the DR11 recall loss is a stage-1 *selection* artifact. This report takes the data/calibration lever to its DR11 limit and reports the resulting v4 candidate set. Two levers, both release-portable (the recipe is what transfers to higher-resolution surveys):

- a calibrated-*mean* stage-1 selector that removes the DR9→DR11 threshold drift at no cost (§2);

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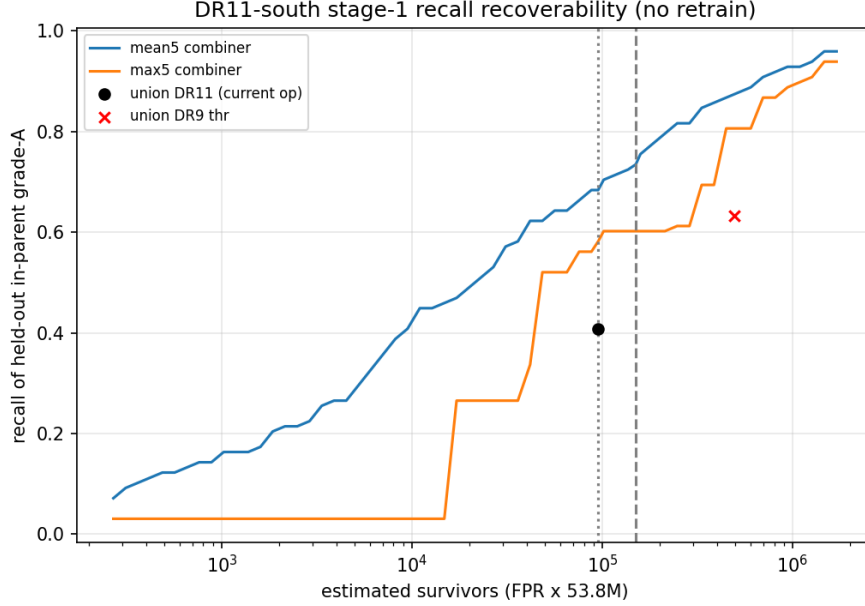


Figure 1: Held-out grade-A recall vs. survivor budget for the mean/max combiners; the per-member union operating point (black) sits far below the mean at the same budget. The recall loss was the selector, not the model.

- a release-native warm-start fine-tune of the three swappable members on a re-harvested, DR11-native training pool (§2), gated on held-out recall.

Both *stack* on *v3blend8*’s contaminant-aware members; *v4* keeps the two frozen anchors (*effnet_B*, *zoobot_N*) and swaps the three fine-tuned members.

2 The v4 ensemble: mean selection + release-native fine-tune

Mean combiner. The stage-1 union over per-member 10^{-4} thresholds degrades on the deeper DR11 imaging (thresholds tighten, the *resnet46_C* member saturates), so the union is dominated by single-member spikes. The calibrated five-member mean instead separates in-parent held-out grade-A lenses from a 2M random DR11 negative sample at AUC 0.9955, recovering grade-A recall 54% $\rightarrow$ 75% at the same 95k budget and reaching DR10’s in-domain 62% at < 43 k survivors (Fig. 1). The mean is adopted as the default stage-1 selector and carries no per-release threshold recalibration.

Release-native training pool + fine-tune. The *v1–v3* positive set was capped at 1,961 DR9-pixel lenses. A broad re-harvest of the confirmed-lens literature for the DR11-south footprint (reclaimed VizieR/DES/KiDS (Rezaei et al., 2025)/HSC catalogs, SLACS/BELLS, lensed-quasar compilations, a bulk SIMBAD pull, Euclid-Q1 A/B (Euclid Collaboration, 2025)), deduplicated at $5''$ and tiered by confidence, yields 3,171 net-new high-confidence positives. All cutouts are re-extracted DR11-native (*griz*, 101 px), tier-subsampled to 5,806 positives, and paired with 30k random + 20k hard DR11 negatives; the three swappable members are warm-started from their *.b50* checkpoints and fine-tuned (12 epochs, $\text{lr}=3 \times 10^{-4}$). Storfer and Inchausti (Inchausti Reyes et al., 2025; Storfer et al., 2024) are held out (excluded as positives *and* negatives), so the gate

Table 1: **v4** fine-tune gate: held-out (training-excluded) recall at the matched-FPR 95k operating point (calibrated-mean combiner). Baseline is the **v3** five-lean ensemble.

held-out set	v3 baseline	v4 fine-tuned	Δ
Inchausti grade-A	0.724	0.816	+0.09
Inchausti grade-B	0.515	0.812	+0.30
Storfer grade-A (hard)	0.544	0.796	+0.25

Table 2: Held-out grade-A recall (position-crossmatch; Storfer/Inchausti training-excluded) across the three DR11-south configurations: the original per-member union operating point, the mean re-rank of the old five-lean ensemble, and the full **v4** re-sweep.

held-out set	union-95k	mean-150k (v3-lean)	v4 re-sweep
Inchausti grade-A	0.54	0.80	0.87 (60/69)
Inchausti grade-B	—	—	0.87 (76/87)
Storfer grade-A (hard)	0.32	0.61	0.825 (85/103)

measures generalisation. The fine-tune passes decisively, with the largest gain on the hard residual (Table 1).

3 The full DR11-south re-sweep and the **v4** candidate set

We re-scored all 5.38×10^7 DR11-south galaxies with the fine-tuned **v4** ensemble — reusing the original parent cutouts (no re-extraction), combining the mean of [**effnet_B**, **zoobot_N**, the three fine-tuned members] — and selected the top 150k by mean. Held-out recall climbs monotonically across the three configurations (Table 2), with the **v4** full re-sweep best on every set and largest on the hard residual. The **v4** top-150k retains only 15,922 of the union-95k survivors and adds 134,078 — a substantially new, recall-rich candidate pool, the input to the vetting below.

4 Candidate catalogue and LensJudge **v3** HSC vetting

In progress. The **v4** candidate set (top-150k by mean; the NEW, not-in-any-catalog shortlist extracted as in the sibling report) is queued for the LENSJUDGE **v3** cascade (DESI tier-1 triage → HSC PDR3 tier-2 escalation). This section will report the HSC-graded A/B yield, the SuGOHI cross-validation, the genuinely-new systems, and the A/B comparison against the **v3blend8** as-run vetted catalogue. Until then the *confirmed* DR11-south catalogue is the **v3blend8** as-run set of the sibling report; the **v4** set here is the recall-rich candidate pool from which the next vetting pass draws.

5 Conclusion

Pending vetting (§4). Provisionally: the **v4** levers — mean selection and a release-native warm-start fine-tune — recover and exceed in-domain DR11-south recall and, for the first time, crack the lrg+companion hard residual, yielding a recall-rich candidate set at the unchanged DECaLS resolution ceiling. Whether this converts to *net-new* confirmed lenses is the question the HSC tier-2 vetting (§4) will answer; the resolution lever (HSC/Euclid) remains decisive.

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